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Progressive uncertainty-guided network for binary segmentation in high-resolution remote sensing imagery

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ABSTRACT

Binary semantic segmentation in remote sensing (RS) imagery faces persistent challenges due to complex object appearances, ambiguous boundaries, and high similarity between foreground and background, all of which introduce significant uncertainty into the prediction process. Existing approaches often treat uncertainty as either a global attribute or a pixel-level estimate, overlooking the critical role of spatial and contextual interactions. To address these limitations, we propose the **Progressive Uncertainty-Guided Segmentation Network (PUGNet)**, a unified framework that explicitly models uncertainty in a context-aware manner. PUGNet decomposes uncertainty into three distinct components: **foreground uncertainty**, **background uncertainty**, and **contextual uncertainty**. This tripartite modeling enables more precise handling of local ambiguities and global inconsistencies. We adopt a coarse-to-fine decoding strategy that progressively refines features through two specialized modules. The **Dynamic Uncertainty-Aware Module** enhances regions of high foreground and background uncertainty using Gaussian-based modeling and contrastive learning. The **Entropy-Driven Refinement Module** quantifies contextual uncertainty via entropy and facilitates adaptive refinement through multi-scale context aggregation. Extensive experiments on ten public benchmark datasets, covering both single-temporal (*e.g.*, building and cropland extraction) and bi-temporal (*e.g.*, building change detection) binary segmentation tasks, demonstrate that PUGNet consistently achieves superior segmentation accuracy and uncertainty reduction, establishing a new state of the art in RS binary segmentation. The full implementation of the proposed framework and all experimental results can be accessed at https://github.com/Henryjiepanli/PU_RS.

1. Introduction

Binary semantic segmentation represents a fundamental task in remote sensing (RS) image analysis, aiming to precisely distinguish foreground objects from background regions at the pixel level. This capability proves indispensable for a multitude of critical applications spanning diverse domains, including building extraction for urban planning and development (Wang et al., 2025a), cropland mapping for precision agriculture and food security monitoring (Liu et al., 2025b), and building change detection for disaster response and infrastructure assessment (Chen et al., 2025). The exponential growth in both the availability and spatial resolution of satellite and aerial imagery (Huang et al., 2024), combined with increasingly sophisticated monitoring requirements (Xiao et al., 2024), has amplified the demand for robust and highly accurate binary segmentation methodologies. Consequently, the development of effective segmentation approaches has emerged as pivotal for advancing large-scale geospatial analysis and facilitating evidence-based decision-making processes. Unlike general saliency segmentation (SS), binary segmentation in RS is semantically-driven: the foreground strictly corresponds to pixels of the target class (*e.g.*, buildings, roads, or crops), while the background includes all remaining pixels. Few-shot segmentation studies in RS (Jia et al., 2025) similarly follow this foreground-background paradigm under extremely limited supervision, highlighting the importance of robust, context-aware feature modeling.

Recent progress in RS binary segmentation has been primarily driven by encoder–decoder frameworks, underpinned by powerful backbones such as ResNet (He et al., 2016), Res2Net (Gao et al., 2019), Swin Transformer (Liu et al., 2021), and Pyramid Vision Transformer (PVT) (Wang et al., 2021c, 2022c). These architectures extract multi-level hierarchical features through deep encoding pipelines, facilitating

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the development of sophisticated decoders that integrate cross-scale information for precise foreground localization.

Decoder architectures have evolved through several distinctive paradigms. Initial endeavors emphasized multi-scale contextual representation via pyramid structures (Lin et al., 2017; Wang et al., 2022b; Pang et al., 2022) and atrous spatial pyramids (Chen et al., 2018; Zhenchao et al., 2023) to capture diverse object appearances and scale variations. Subsequent works incorporated global attention mechanisms, including self-attention (Guo et al., 2024; Xu et al., 2023b) and transformer-based modules (Wang et al., 2022a; Xu et al., 2023a), to enhance semantic coherence through comprehensive long-range dependency modeling. Contemporary strategies prioritize fine-grained detail recovery through edge-aware supervision (Guo et al., 2022; Wang et al., 2023), frequency-domain enhancement (Li et al., 2025e; Gao et al., 2025), and coarse-to-fine refinement approaches (Li et al., 2024b, 2025a; Guo et al., 2022; Chen et al., 2024b; Pang et al., 2024), wherein initial predictions undergo progressive refinement to reconstruct intricate boundary details.

Despite these substantial architectural advancements, binary segmentation in RS imagery remains fundamentally constrained by pervasive uncertainty stemming from diverse object scales, ambiguous boundaries, and pronounced foreground-background similarity. These uncertainties not only compromise segmentation accuracy but also undermine the reliability of subsequent geospatial analysis workflows. Contemporary uncertainty modeling approaches (Kendall and Gal, 2017) can be systematically categorized into two principal paradigms: (1) Aleatoric uncertainty, which captures data-inherent noise through probabilistic models such as Bayesian neural networks (Dechesne et al., 2021; Celikkan et al., 2023) and conditional VAEs (Li et al., 2025a; Zhang et al., 2023; Wu et al., 2025); (2) Epistemic uncertainty, which quantifies model knowledge limitations through confidence estimation techniques (Li et al., 2024b; Liu et al., 2025a) and uncertainty-aware loss formulations (Wang et al., 2021b; Xu et al., 2024).

While both paradigms have demonstrated notable effectiveness, they share a common limitation in that **uncertainty is typically modeled in a largely context-independent manner**. Aleatoric methods assume a global data distribution, and epistemic approaches focus on pixel-level confidence without explicitly modeling interactions between neighboring regions. In RS imagery, semantic interpretation is highly context-sensitive; for example, the uncertainty in segmenting a building pixel depends not only on its spectral signature but also on its spatial relationships with adjacent elements such as vegetation, roads, or other structures. Neglecting such local object-context interactions (Li et al., 2025b; Wang et al., 2025b) can lead to fragmented segmentation and unreliable uncertainty estimates.

To overcome the aforementioned limitations, we introduce the Progressive Uncertainty-Guided Segmentation Network (PUGNet), a unified and context-aware framework designed to enhance binary semantic segmentation in RS imagery by explicitly modeling spatially adaptive uncertainty. The core insight of PUGNet is a tripartite decomposition of uncertainty into three complementary components: foreground uncertainty (arising from ambiguous object regions), background uncertainty (linked to cluttered or low-contrast non-target areas), and contextual uncertainty (centered around object boundaries and transition zones). Such a decomposition is crucial because different types of uncertainty, including object-intrinsic, background-induced, and context-dependent uncertainty, affect predictions in different ways. Modeling them with a single, undifferentiated representation may obscure their individual roles, limiting the effectiveness of refinement in critical regions. By explicitly modeling each component, PUGNet provides fine-grained guidance, allowing targeted refinement of ambiguous regions and yielding a more interpretable and accurate. As illustrated in Fig. 1, this decomposition enables the model to distinguish between local ambiguities such as object-to-background transitions and broader inconsistencies in global prediction confidence, thereby supporting a more fine-grained and interpretable uncertainty representation.

In detail, our PUGNet introduces a coarse-to-fine framework designed to address uncertainty through progressive refinement across multiple levels. Starting with coarse predictions derived from high-level features, uncertainty is explicitly decoupled into three categories: foreground, background, and contextual uncertainty. To handle foreground and background uncertainties, we propose a **Dynamic Uncertainty-Aware Module**, which initializes uncertainty centers and models uncertainty using Gaussian distributions. This module highlights high-uncertainty regions, enabling targeted enhancement. Additionally, contrastive learning is employed to maximize the distinction between enhanced foreground and background features before they are fused, further improving segmentation accuracy. For contextual uncertainty, we incorporate an **Entropy-Driven Refinement Module**, which quantifies uncertainty using entropy and captures contextual information across multiple scales. This module allows the network to adaptively focus on complex regions. To ensure stability, a Monte Carlo-based strategy is seamlessly integrated within the module, addressing inconsistencies and refining feature representations for greater precision. By progressively guiding feature refinement through uncertainty-aware mechanisms, PUGNet achieves accurate and consistent segmentation with reduced uncertainty, effectively handling complex scenarios and challenging environments.

Our main contributions are summarized as follows:

- **Tripartite Context-Aware Uncertainty Modeling**: We propose a novel decomposition of segmentation uncertainty into foreground, background, and contextual components, explicitly modeling localized object-context interactions and global consistency in RS scenes.
- **Progressive Uncertainty Integration**: We design a multi-stage, coarse-to-fine framework that integrates uncertainty cues through specialized modules (DUAM and EDRM), enabling adaptive guidance throughout the segmentation pipeline.
- **Comprehensive Benchmark Evaluation Across Tasks**: Extensive experiments on ten public benchmark datasets, encompassing single-temporal tasks (*e.g.*, building and cropland extraction) and bi-temporal tasks (*e.g.*, building change detection), demonstrate that PUGNet consistently achieves state-of-the-art performance across a wide range of RS scenarios.

2. Methodology

2.1. Theoretical foundation

We model predictive uncertainty in RS segmentation via a Region-Aware Bayesian Mixture, explicitly accounting for foreground (f), background (b), and contextual transition (c) uncertainties.

Let θ denote the latent model parameters, x denote the input RS image, and y denote the corresponding ground-truth segmentation map. The standard Bayesian prediction is formulated as:

$$p(y \mid x) = \int p(y \mid x, \theta)\, p(\theta \mid x)\, d\theta.$$

To capture the semantic and spatial heterogeneity inherent in RS data, we introduce a discrete latent variable $z \in \{f, b, c\}$ indicating the semantic region type. The joint posterior factorizes as $p(\theta, z|x) = p(\theta|x, z)\, p(z|x)$, leading to the predictive mixture

$$p(y|x) = \sum_{z\in\{f,b,c\}} p(z \mid x) \int p(y|x, \theta)\, p(\theta|x, z)\, d\theta. \tag{1}$$

Marginalizing z, the feature posterior becomes a region-aware mixture:

$$p(\theta \mid x) = \alpha\, p_f(\theta \mid x) \;+\; \beta\, p_b(\theta \mid x) \;+\; \gamma\, p_c(\theta \mid x), \tag{2}$$

where α, β, and γ are pixel-wise adaptive coefficients derived from the coarse segmentation output. These coefficients function as region-sensitive weights that modulate the contributions of the foreground-, background-, and transition-specific uncertainty distributions, and

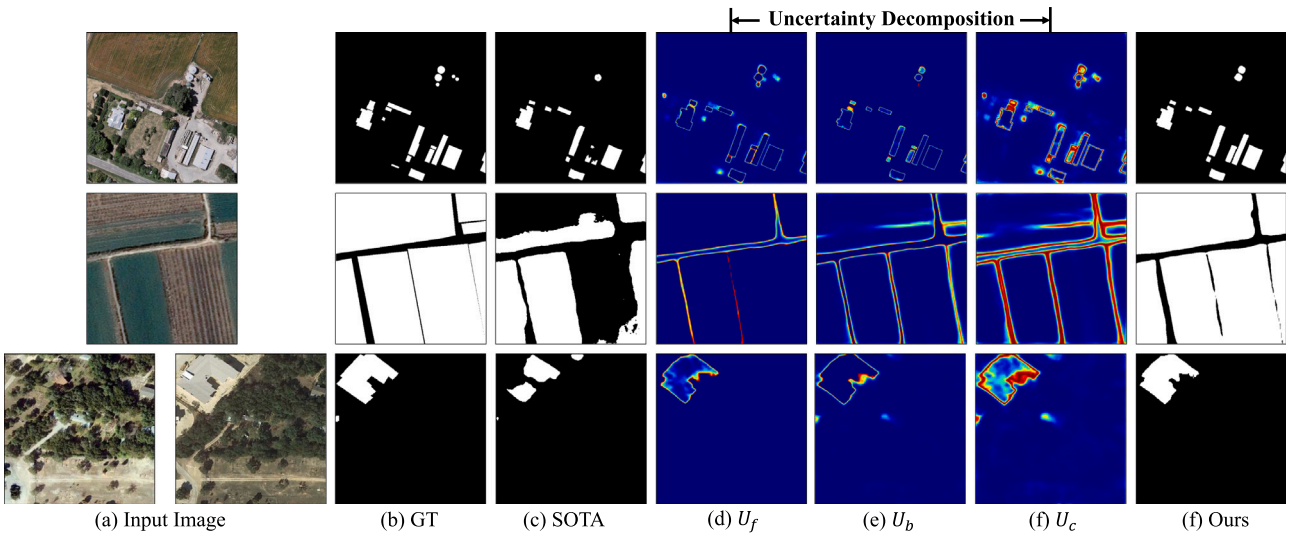


Fig. 1. Comparison with state-of-the-art (SOTA) binary segmentation methods, including HDNet (Li et al., 2024c) for building extraction, DBBANet (Li et al., 2025d) for cropland mapping, and BAN (Li et al., 2024a) for building change detection, along with visualizations of our proposed uncertainty decomposition. The components U_f, U_b, and U_c represent the estimated uncertainty associated with the foreground, background, and contextual regions, respectively.

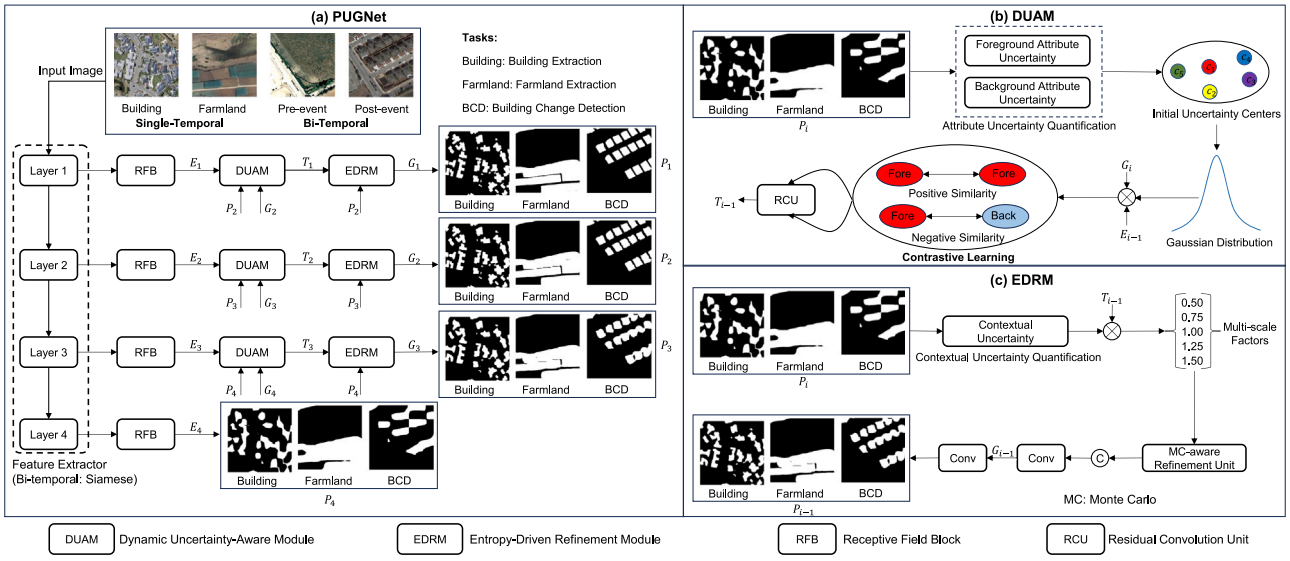


Fig. 2. Overall structure of the proposed PUGNet and details of its core modules. (a) The complete PUGNet framework, showing the coarse-to-fine feature flow and multi-stage integration of uncertainty maps. (b) Detailed structure of the Dynamic Uncertainty-Aware Module (DUAM), which leverages foreground and background uncertainty to guide feature refinement. (c) Detailed structure of the Entropy-Driven Refinement Module (EDRM), which incorporates contextual uncertainty to enhance the stability and consistency of segmentation outputs. Arrows indicate the flow of features and uncertainty maps between modules and stages.

are implicitly refined during end-to-end training without requiring additional supervision.

Necessity of the Decomposition: In RS segmentation, predictive uncertainty arises from heterogeneous sources that are strongly coupled with spatial semantics. Conditioning θ on the region type z allows different modes of uncertainty to be explicitly modeled:

- $p_f(\theta|x)$ captures foreground uncertainty, mainly reflecting omission ambiguity and incomplete object activation within target regions;
- $p_b(\theta|x)$ models background uncertainty, corresponding to spurious responses caused by background clutter or foreground-like textures;
- $p_c(\theta|x)$ represents contextual uncertainty, which dominates in boundary areas and mixed pixels where local spatial context is ambiguous.

Sufficiency for Segmentation: This decomposition provides a semantically and spatially complete view of uncertainty, enabling the model to suppress noise, sharpen boundaries, and generalize effectively across complex RS scenes.

2.2. Overview of the proposed framework

As illustrated in Fig. 2, our framework supports both single-temporal and bi-temporal segmentation tasks. For single-temporal tasks (*e.g.*, building or cropland extraction), the input image I is passed through a standard encoder (PVT-v2 Wang et al., 2022c) to extract multi-level features. For bi-temporal tasks (*e.g.*, building change detection), a Siamese encoder processes pre- and post-event images independently, followed by channel-wise concatenation and a 1×1 convolution for feature alignment.

In the decoding stage, we first apply the Receptive Field Block (RFB) (Liu et al., 2018) to capture multi-scale context from each level of encoded features $E_i \in \mathbb{R}^{B\times C_i\times H_i\times W_i}$, where $H_i = \frac{H}{2^{i+1}}$, $W_i = \frac{W}{2^{i+1}}$. A coarse prediction P_4 is obtained by applying a convolution to the top-level features E_4. To explicitly address uncertainty, we decompose P_4 into three complementary types: **foreground**, **background**, and **contextual uncertainties**. A Dynamic Uncertainty-Aware Module (DUAM) is then employed to refine the fusion of adjacent features E_4 and E_3, alleviating the influence of ambiguous regions and enhancing foreground–background separation. To further improve contextual consistency, we introduce an Entropy-Driven Refinement Module (EDRM), which utilizes contextual uncertainty to generate a refined prediction P_3. Meanwhile, EDRM also produces an intermediate fused feature G_3, which serves as the input representation for the subsequent refinement stage. Finally, we adopt a progressive prediction-guided refinement strategy. Specifically, the prediction P_3 is used to guide the fusion of G_3 and E_2, yielding the refined prediction P_2 and its corresponding fused feature G_2. Similarly, P_2 guides the fusion of G_2 and E_1 to produce the final prediction P_1.

2.3. Dynamic uncertainty-aware module

As illustrated in Fig. 2(b), given the coarse segmentation outputs P_i, $i \in \{2,3,4\}$, we directly quantify both **foreground uncertainty** U_i^f and **background uncertainty** U_i^b as follows:

$$\begin{aligned} U_i^f &= \max\big(\sigma(P_i) - 0.5,\, 0\big),\\ U_i^b &= \max\big(0.5 - \sigma(P_i),\, 0\big), \end{aligned} \tag{3}$$

where $\sigma(\cdot)$ denotes the Sigmoid activation function.

Considering the inherently distinct relationships between foreground and background in binary segmentation, it is crucial to treat their uncertainties separately. Based on the computed foreground and background uncertainties, we propose a Gaussian soft ranking algorithm to locally assess feature uncertainty from a spatial perspective. This mechanism dynamically highlights features according to their uncertainty levels.

Specifically, we define a set of learnable Gaussian centers $\{c_i\}_{i=1}^N$ (with $N = 5$ in our implementation). For each pixel located at (x, y), the ranking score with respect to the ith Gaussian center is computed by:

$$R_i(x,y) = \exp\left(-\frac{\big(U_i(x,y) - c_i\big)^2}{2\beta^2}\right), \tag{4}$$

where β controls the smoothness of the Gaussian kernel, thus regulating the sensitivity of the ranking to uncertainty deviations. Here, $U_i(x,y)$ corresponds to the uncertainty value at pixel (x,y), either $U_i^f(x,y)$ for foreground or $U_i^b(x,y)$ for background.

The overall ranking for each pixel is obtained by aggregating the contributions from all Gaussian centers and normalizing:

$$R_i^{\text{norm}}(x,y) = \frac{R_i(x,y)}{\sum_{i=1}^N R_i(x,y)}. \tag{5}$$

This Gaussian soft ranking mechanism enables a fine-grained, adaptive emphasis on regions with higher uncertainty, effectively guiding subsequent feature refinement.

Given input features F (which can be either E_{i-1} or G_i in our framework), the refined features are obtained by weighting the original feature map with the normalized uncertainty rankings. Formally, the weighted foreground and background features are computed as:

$$\begin{aligned} F_w^f &= F \odot \sum_{i=1}^N R_i^{\text{norm},f}(x,y),\\ F_w^b &= F \odot \sum_{i=1}^N R_i^{\text{norm},b}(x,y), \end{aligned} \tag{6}$$

where $\odot$ denotes element-wise multiplication, and the summation is performed over the Gaussian centers.

To explicitly enhance the discriminability between foreground and background feature representations, we employ a contrastive learning strategy, which enforces the model to learn more distinct feature embeddings based on their uncertainty-aware weighting.

Given a batch of weighted foreground features $F_w^f \in \mathbb{R}^{B\times C\times H\times W}$ and background features $F_w^b \in \mathbb{R}^{B\times C\times H\times W}$, we first flatten the spatial and channel dimensions and apply L_2 normalization:

$$\begin{aligned} \hat{F}_f &= \text{normalize}\big(F_w^f.\text{view}(B,-1)\big),\\ \hat{F}_b &= \text{normalize}\big(F_w^b.\text{view}(B,-1)\big), \end{aligned} \tag{7}$$

where $\text{normalize}(\cdot)$ denotes normalization along the feature dimension.

We then compute the cosine similarity matrices between foreground and background features:

$$\mathbf{S}_{\text{neg}} = \hat{F}_f \hat{F}_b^\top \in \mathbb{R}^{B\times B}, \tag{8}$$

and the positive similarity vector (self-similarity of foreground features):

$$\mathbf{S}_{\text{pos}} = \text{diag}\big(\hat{F}_f \hat{F}_f^\top\big) \in \mathbb{R}^B. \tag{9}$$

To focus training on the most challenging negative pairs, we select the hardest negative similarity for each sample:

$$\mathbf{S}_{\text{neg}}^{\text{hard}}[i] = \max_j \mathbf{S}_{\text{neg}}[i,j], \quad i = 1,\dots,B. \tag{10}$$

The contrastive loss is formulated as a cross-entropy objective encouraging the model to maximize the similarity of positive pairs relative to the negatives:

$$\mathcal{L}_c = -\frac{1}{B}\sum_{i=1}^B \log \frac{\exp\big(\mathbf{S}_{\text{pos}}[i]/\tau\big)}{\exp\big(\mathbf{S}_{\text{pos}}[i]/\tau\big) + \sum_{j=1}^B \exp\big(\mathbf{S}_{\text{neg}}[i,j]/\tau\big)}, \tag{11}$$

where τ is a temperature hyperparameter that controls the concentration of the distribution.

This loss effectively pulls together features corresponding to foreground regions while pushing apart features between foreground and background, thereby improving the model's ability to distinguish regions with varying uncertainty.

Finally, the weighted foreground and background features are concatenated along the channel dimension and passed through a 1×1 convolution to obtain the fused representation O. With the dynamic uncertainty-aware module (DUAM), adjacent features are progressively refined under the guidance of the coarse segmentation maps $P_i, i \in \{2,3,4\}$. The refinement process is formulated as:

$$\begin{aligned} O_i &= \text{DUAM}(E_i, P_i),\\ O_{i-1} &= \text{DUAM}(E_{i-1}, U^2(P_i)),\\ T_{i-1} &= \text{RCU}\big(U^2(O_i) + \text{RCU}(O_{i-1})\big), \end{aligned} \tag{12}$$

where U^2 denotes upsampling by a factor of 2, and $\text{RCU}(\cdot)$ is the Residual Convolution Unit defined as:

$$\text{RCU}(X) = X + \phi\big(\text{Conv}_{3\times3}\big(\phi(\text{Conv}_{3\times3}(X))\big)\big), \tag{13}$$

with ϕ representing the activation function.

This module thus dynamically leverages uncertainty information to guide feature refinement, leading to improved segmentation accuracy, especially in ambiguous regions.

2.4. Entropy-driven refinement module

As illustrated in Fig. 2(c), to handle the **contextual uncertainty** related to the stability and consistency of segmentation outputs, we propose an entropy-driven refinement module.

The refinement begins by upsampling the coarse prediction P_{i+1} from the $(i+1)^{\text{th}}$ layer to match the spatial resolution of the corresponding feature map T_i. From the upsampled probability map, we derive an

entropy-based uncertainty map to quantify prediction confidence. The entropy map is subsequently normalized to the range $[0, 1]$ to yield the contextual guidance map M_i. This procedure is formulated as:

$$\begin{aligned} \tilde{P}_i &= \sigma\left(U^2(P_{i+1})\right), \\ M_i &= \sigma\left(-\tilde{P}_i \log \tilde{P}_i - (1-\tilde{P}_i)\log(1-\tilde{P}_i)\right), \end{aligned} \tag{14}$$

where $\tilde{P}_i$ denotes the upsampled probability map, U^2 represents bilinear upsampling by a factor of 2, and $\sigma(\cdot)$ is the sigmoid function.

To enhance global contextual understanding, we incorporate a multi-scale processing scheme. Specifically, the features T_i and the guidance map M_i are resized by scale factors $S_j \in \{S_1, S_2, \ldots, S_n\}$, where $j = 1, 2, \ldots, n$. At each scale, the resized features and uncertainty maps are fused via element-wise multiplication to emphasize high-uncertainty regions. Then, convolutional operations with varying kernel sizes k_j are applied in parallel to extract scale-specific semantic features. These are aggregated using learnable weights α_j and refined via a Squeeze-and-Excitation (SE) attention module (Hu et al., 2018), as follows:

$$\begin{aligned} &T_i^{(j)} = U^{S_j}(T_i), \quad M_i^{(j)} = U^{S_j}(M_i), \\ &\hat{F}_i^{(j)} = \mathrm{SE}\left(\sum_{j=1}^{n} \alpha_j \cdot \mathrm{Conv}_{k_j}\left(T_i^{(j)} \odot M_i^{(j)}\right)\right), \end{aligned} \tag{15}$$

where $\odot$ denotes element-wise multiplication, $\mathrm{Conv}_{k_j}(\cdot)$ is a convolution with kernel size k_j, and α_j are learnable weights that control the contribution of each scale.

To further capture epistemic uncertainty, we introduce a Monte Carlo sampling-based refinement. We perform N stochastic forward passes to estimate the mean and variance of the feature map $\hat{F}_i^{(j)}$, thereby modeling uncertainty through distributional spread:

$$\begin{aligned} \mu_i^{(j)} &= \frac{1}{N}\sum_{t=1}^{N} \hat{F}_i^{(j,t)}, \\ \sigma_i^{2(j)} &= \frac{1}{N}\sum_{t=1}^{N} \left(\hat{F}_i^{(j,t)} - \mu_i^{(j)}\right)^2, \end{aligned} \tag{16}$$

where $\hat{F}_i^{(j,t)}$ denotes the feature map at scale j obtained during the tth stochastic pass.

The final uncertainty-aware refined feature map $\tilde{F}_i^{(j)}$ is computed as:

$$\tilde{F}_i^{(j)} = \mu_i^{(j)} + \sigma\left(\sigma_i^{2(j)}\right) \odot \hat{F}_i^{(j)}, \tag{17}$$

where the variance is normalized by the sigmoid function to ensure numerical stability and boundedness.

All scale-specific refined features $\tilde{F}_i^{(j)}$ are then resized to the original resolution of T_i and concatenated along the channel axis:

$$F_i^{\mathrm{cat}} = \mathrm{Concat}\left(U^{1/S_j}\left(\tilde{F}_i^{(j)}\right)\right)_{j=1}^{n}. \tag{18}$$

Finally, F_i^{cat} is passed through a convolutional fusion layer to obtain the enhanced feature representation G_i, which directly predicts the refined segmentation map P_i. This integration step is crucial, as it combines the entropy-guided contextual cues with the Monte Carlo uncertainty estimate to adaptively refine predictions in ambiguous regions.

2.5. Loss function

As illustrated in Fig. 2, the proposed model generates four hierarchical segmentation maps, denoted as P_1, P_2, P_3, and P_4, each of which undergoes direct supervision against the ground truth (GT). For each prediction level, we employ the cross-entropy loss ($\mathcal{L}_{\mathrm{ce}}$) to enforce segmentation supervision and ensure consistency across the hierarchical outputs. Furthermore, our Dynamic Uncertainty-Aware Module incorporates a contrastive learning strategy to enhance feature discriminability between foreground and background regions. Specifically, the module introduces three complementary contrastive loss terms, denoted as $\mathcal{L}^i_{\mathrm{contrastive}}$ for $i = 1, 2, 3$, which are strategically designed to maximize the separation between foreground and background representations in the learned feature space. Consequently, the overall loss function is formulated as:

$$\mathcal{L}_{\mathrm{total}} = \sum_{i=1}^{4} \mathcal{L}_{\mathrm{ce}}(P_i, GT) + \sum_{i=1}^{3} \mathcal{L}^i_{\mathrm{contrastive}}. \tag{19}$$

This comprehensive joint supervision scheme enables the model to simultaneously generate accurate pixel-wise predictions and learn uncertainty-aware, discriminative representations, thereby substantially improving segmentation performance in semantically challenging and ambiguous regions.

3. Experiment

3.1. Datasets

To comprehensively evaluate the effectiveness of the proposed PUGNet, we conduct extensive experiments on a diverse set of RS binary segmentation tasks, including both single-temporal (*e.g.*, building and cropland extraction) and bi-temporal scenarios (*e.g.*, building change detection).

For the building extraction task, we utilize three widely-used public datasets: the WHU Building Dataset (Ji et al., 2018a), the Massachusetts Building Dataset (Mnih, 2013), and the Inria Aerial Image Labeling Dataset (Maggiori et al., 2017). For cropland extraction, we adopt the FGFD dataset (Li et al., 2025d), which provides high-resolution images with detailed cropland annotations. For bi-temporal building change detection, we evaluate our method on three benchmark datasets: LEVIR-CD (Chen and Shi, 2020), SYSU-CD (Shi et al., 2021), Lebedev dataset (Jiang et al., 2020), WHU-CD dataset (Ji et al., 2018b), Google-CD dataset (Peng et al., 2020), and the LEVIR-CD+ dataset (Chen and Shi, 2020), each offering paired images captured at different times along with corresponding change labels.

A detailed summary of these datasets, including spatial resolution, number of images, and annotation types, is provided in Table 1.

3.2. Implementation details

To ensure fair and consistent comparisons with SOTA binary segmentation methods across various tasks, we adopt task-specific preprocessing strategies aligned with prior work. Specifically, for building extraction, we follow the preprocessing pipeline of BuildFormer (Wang et al., 2022a); for cropland extraction, we follow the strategy used in DBBANet (Li et al., 2025d); and for building change detection, we adopt the same preprocessing approach as CGNet (Han et al., 2023a).

All experiments are conducted using PyTorch 1.10 and Python 3.8 on an NVIDIA GeForce RTX 4090 GPU with 24 GB of memory. During training, we employ standard online data augmentation techniques, including random horizontal and vertical flipping, random rotation, and mosaic augmentation. For the building and cropland extraction tasks, we utilize the AdamW optimizer and adopt a cosine annealing learning rate schedule. The initial learning rate is set to 5×10^{-4}, and training proceeds for 105 epochs. In the case of building change detection, we employ the Adam optimizer with an initial learning rate of 1×10^{-4}. The learning rate is decayed by a factor of 10 every 50 epochs, and the total number of training epochs is 200.

In addition, several task-independent hyperparameters in our method are fixed across all experiments for reproducibility. Specifically, the parameter β controlling the smoothness of the Gaussian kernel in the uncertainty ranking module is set to 0.1; the temperature parameter τ in the contrastive formulation is set to 0.07; and the multi-scale convolution kernels $\mathrm{Conv}_{k_j}(\cdot)$ adopt kernel sizes $k_j \in \{3, 5, 7\}$.

Table 1
Summary of tasks and the corresponding datasets used in our experiments.

Dataset	Resolution	Image size	Region	Split (Train/Val/Test)
Single-temporal building extraction				
WHU building dataset	0.3 m	512 × 512	Christchurch, New Zealand	4736/1036/2416
Massachusetts building dataset	1 m	1500 × 1500	Boston, USA	137/4/10
Inria aerial image labeling dataset	0.3 m	5000 × 5000	Austin, Chicago, Kitsap, Tyrol, Vienna	155/–/25
Single-temporal cropland extraction				
Fine-grained farmland dataset	0.3 m	512 × 512	China	2083/263/260
Bi-temporal building change detection				
LEVIR-CD dataset	0.5 m	1024 × 1024	Texas, USA	455/64/128
SYSU-CD dataset	0.5 m	256 × 256	Hong Kong, China	12000/4000/4000
Lebedev dataset	0.1–0.3 m	256 × 256	–	10000/2998/3000
WHU-CD dataset	0.3 m	256 × 256	Christchurch, New Zealand	4536/504/2760
Google-CD dataset	0.5 m	256 × 256	Guangzhou, China	740/116/300
LEVIR-CD+	0.5 m	1024 × 1024	–	637/-/348

Table 2
Performance comparison of single-temporal building extraction with baseline models on the WHU, Massachusetts, and Inria datasets. ↑ indicates the higher score the better and vice versa. The best score for each metric is marked in bold. Note that for SAM and SAM2, we use the test predictions generated by our PUGNet as prompts, while for SAM3, we use the textual prompt "building".

Baseline	WHU (%)				Massachusetts (%)				Inira (%)			
	IoU ↑	*F1* ↑	*Pre* ↑	*Recall* ↑	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
CNN-based backbone												
UNet	85.92	92.39	91.78	93.01	68.48	81.47	80.99	81.96	74.40	85.32	86.39	84.28
HRNet	85.64	92.27	91.69	92.85	69.39	81.93	81.49	82.38	75.03	85.73	86.56	84.92
DSNet	89.54	94.48	94.05	94.91	75.04	85.74	87.56	83.99	81.02	89.52	90.32	88.73
CBRNet	91.40	95.51	95.31	**95.70**	74.55	85.42	86.50	84.36	81.10	89.56	89.93	89.20
HDNet	90.19	94.84	95.00	94.68	75.53	86.06	85.98	**86.13**	82.10	90.17	91.10	89.26
FD-Net	91.25	95.39	95.31	95.48	74.43	85.37	87.88	83.00	81.38	89.74	90.67	88.82
Transformer-based backbone												
BuildFormer	90.73	95.14	95.15	95.14	75.03	85.73	86.69	84.79	81.24	89.71	90.65	88.78
BCTNet	91.15	95.37	95.47	95.27	75.04	85.74	87.57	83.99	81.45	89.78	90.72	88.85
UANet	91.68	95.66	95.89	95.44	75.84	86.25	87.41	85.12	83.34	90.91	91.86	89.97
MSHFormer	90.15	95.26	95.93	94.60	74.23	85.26	86.38	84.17	81.04	89.57	90.45	88.70
SAM	85.65	92.27	92.11	92.43	62.63	77.02	77.32	76.72	71.17	83.16	82.76	83.56
SAM2	83.77	91.17	90.47	91.88	61.94	76.50	77.03	75.98	72.09	83.78	83.52	84.04
SAM3	84.37	91.52	89.83	93.29	37.04	54.05	82.53	40.19	69.96	82.32	83.85	80.85
PUGNet	**91.78**	**95.71**	**96.00**	95.43	**76.29**	**86.55**	**88.87**	84.35	**83.93**	**91.26**	**92.53**	**90.03**

Table 3
Performance comparison of single-temporal cropland extraction with baseline models on the FGFD dataset. ↑ indicates that higher scores are better. The best score for each metric is marked in bold. Note that for SAM and SAM2, we use the test predictions generated by our PUGNet as prompts, while for SAM3, we use the textual prompt "cropland".

Baseline	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
CNN-based backbone				
HRNet	82.45	90.88	90.92	91.04
DenseASPP	82.72	90.54	90.15	89.94
UNet	82.83	90.61	88.70	92.60
MCCANet	85.30	92.07	91.83	92.31
CMTFNet	85.01	91.90	91.04	92.77
DBBANet	86.04	92.50	91.81	93.19
Transformer-based backbone				
BuildFormer	84.25	91.45	90.08	92.03
UNetFormer	84.58	91.65	91.54	91.75
UANet	85.69	93.00	92.13	92.46
AAFormer	85.80	92.36	91.39	93.34
SAM	83.34	90.91	90.21	91.63
SAM2	83.31	90.90	90.21	91.60
SAM3	10.98	19.79	72.02	11.47
PUGNet	**88.66**	**93.99**	**92.32**	**95.72**

3.3. Evaluation metrics

To comprehensively assess the performance of our proposed model, we adopt four widely-used evaluation metrics: Intersection over Union (*IoU*), F1 score (*F1*), Precision (*Pre*), and Recall (*Rec*). Let TP, FP, and FN denote the number of true positives, false positives, and false negatives, respectively. The definitions of the evaluation metrics are as follows:

$$IoU = \frac{TP}{TP + FP + FN}, \tag{20}$$

$$Precision = \frac{TP}{TP + FP}, \tag{21}$$

$$Recall = \frac{TP}{TP + FN}, \tag{22}$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}. \tag{23}$$

3.4. Compared methods

For single-temporal building extraction, we evaluate a diverse set of representative approaches. The CNN-based models include UNet (Ronneberger et al., 2015), HRNet (Wang et al., 2020), DSNet (Zhang et al., 2020), CBRNet (Guo et al., 2022), HDNet (Li et al., 2024c), and FD-Net (Guo et al., 2023). We further incorporate Transformer-based architectures such as BuildFormer (Wang et al., 2022a), BCTNet (Xu

Table 4
Performance comparison of bi-temporal building change detection on the LEVIR-CD, SYSU-CD, Lebedev, WHU-CD, Google-CD, and LEVIR-CD+ datasets. ↑ indicates the higher score the better and vice versa. The best score for each metric is marked in bold.

Baseline	LEVIR-CD (%)				SYSU-CD (%)				Lebedev (%)			
	IoU ↑	*F1* ↑	*Pre* ↑	*Recall* ↑	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
CNN-based backbone												
FC-Siam-conc	82.24	90.25	89.67	90.84	66.16	79.64	83.27	76.31	84.28	91.47	88.72	94.40
FC-Siam-diff	82.24	90.26	90.40	90.11	66.93	80.19	79.96	80.42	85.94	92.44	89.99	95.02
SNUNet	82.65	90.50	90.46	90.54	64.79	78.63	80.88	76.50	90.65	95.10	91.55	98.93
P2V-CD	83.67	91.11	91.01	91.21	64.35	78.31	81.93	74.99	91.14	95.36	91.51	**99.56**
HANet	82.27	90.28	91.21	89.36	63.14	77.41	78.71	76.14	77.51	87.33	92.33	82.85
CGNet	85.21	92.01	93.15	90.90	66.55	79.92	86.37	74.37	91.41	95.51	96.16	94.48
Mamba-based backbone												
RS-Mamba	82.07	90.15	91.15	89.18	65.49	79.16	81.48	76.97	90.62	95.07	91.46	98.74
ChangeMamba	85.20	92.01	93.01	91.03	70.00	82.35	88.16	77.27	93.43	96.61	96.65	96.56
Transformer-based backbone												
BIT	82.76	90.56	90.87	90.26	65.33	79.03	78.23	79.84	89.19	94.29	91.14	97.66
ChangeFormer	81.25	89.65	89.70	89.61	65.24	78.97	81.21	76.84	89.58	94.50	90.93	98.37
M-Swin	83.58	91.05	92.08	90.05	66.44	79.90	87.46	73.55	91.84	95.75	96.12	95.38
BAN	84.93	91.85	92.93	90.89	67.35	80.49	**88.26**	73.97	88.47	93.88	94.54	93.23
UA-BCD	85.99	92.47	93.38	**91.57**	71.38	83.30	87.28	79.66	93.49	96.64	96.38	96.90
PUGNet (ours)	**86.03**	**92.49**	**93.59**	91.42	**72.30**	**83.92**	88.15	**80.08**	**94.00**	**96.91**	**96.88**	96.93
Baseline	WHU-CD (%)				Google-CD (%)				LEVIR-CD+ (%)			
	IoU ↑	*F1* ↑	*Pre* ↑	*Recall* ↑	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
CNN-based backbone												
FC-Siam-conc	40.89	58.05	76.49	46.77	71.12	83.13	79.86	86.67	64.53	78.44	78.39	78.49
FC-Siam-diff	47.05	63.99	72.06	57.55	71.52	83.40	82.72	84.08	63.36	77.57	81.49	74.02
SNUNet	78.19	87.76	87.84	87.68	72.97	84.37	83.65	85.11	59.62	74.70	71.07	78.73
P2V-CD	78.03	87.66	86.99	88.33	74.04	85.08	87.12	83.14	71.86	83.52	81.29	85.88
HANet	78.82	88.16	88.30	88.01	73.14	84.49	86.03	83.00	63.34	77.56	79.70	75.53
CGNet	86.21	92.59	94.47	90.79	75.27	85.89	88.07	83.82	71.94	83.68	81.46	86.02
Mamba-based backbone												
RS-Mamba	85.60	92.29	93.66	90.95	74.33	85.30	87.39	83.31	72.08	83.77	81.57	86.10
ChangeMamba	88.13	93.69	96.17	91.33	79.21	88.40	89.00	87.81	78.59	87.94	88.69	87.21
Transformer-based backbone												
BIT	68.02	80.97	74.01	89.37	75.73	86.19	86.57	85.81	71.44	83.34	85.87	80.96
ChangeFormer	77.27	87.16	92.70	82.28	71.90	83.65	85.18	82.18	67.58	80.65	81.34	79.97
M-Swin	80.85	89.41	90.39	88.45	76.12	86.22	88.17	84.35	69.74	82.03	83.50	80.61
BAN	86.05	92.57	94.03	91.15	78.70	88.11	89.22	87.02	76.86	86.98	87.55	86.42
UA-BCD	86.87	92.98	94.71	91.30	79.86	88.81	89.66	**87.97**	79.11	88.33	89.16	87.52
PUGNet (ours)	**88.86**	**94.10**	**96.65**	**91.69**	**80.61**	**89.26**	**91.02**	87.57	**79.58**	**88.55**	**89.39**	**87.72**

et al., 2023a), UANet (Li et al., 2024b), and MSHFormer (Zhu et al., 2025). In addition, we compare our method with the Segment Anything Model family, including SAM (Kirillov et al., 2023), SAM2 (Ravi et al., 2024) and SAM3 (Carion et al., 2025).

For single-temporal cropland extraction, we evaluate a set of representative CNN-based models, including HRNet (Wang et al., 2020), DenseASPP (Yang et al., 2018), UNet (Ronneberger et al., 2015), and DBBANet (Li et al., 2025d). In parallel, we consider several Transformer-based approaches, such as BuildFormer (Wang et al., 2022a), UNetFormer (Wang et al., 2022b), CMTFNet (Wu et al., 2023), MCCANet (Zheng et al., 2023), UANet (Li et al., 2024b), and AAFormer (Li et al., 2024d). Furthermore, we compare our method against the Segment Anything Model family, including SAM (Kirillov et al., 2023), SAM2 (Ravi et al., 2024), and SAM3 (Carion et al., 2025).

For bi-temporal building change detection, we compare our PUGNet with a broad spectrum of state-of-the-art approaches. The CNN-based baselines include FC-Siam-conc (Daudt et al., 2018), FC-Siam-diff (Daudt et al., 2018), SNUNet (Chen et al., 2022), P2V-CD (Lin et al., 2022), HANet (Han et al., 2023b), and CGNet (Han et al., 2023a). The Transformer-based counterparts include BIT (Chen et al., 2021), ChangeFormer (Bandara and Patel, 2022), M-Swin Pan et al. (2024), BAN (Li et al., 2024a), and UA-BCD (Li et al., 2025a). In addition, we further compare against two recent Mamba-based models, RS-Mamba (Zhao et al., 2024) and ChangeMamba (Chen et al., 2024a), to provide a comprehensive evaluation.

3.5. Quantitative evaluation

3.5.1. Single-temporal building extraction

As reported in Table 2, our proposed PUGNet achieves the best performance across all metrics on the three benchmark datasets. On the WHU Building Dataset, PUGNet surpasses the SOTA method UANet (Li et al., 2024b) by 0.10% in IoU, 0.05% in F1-score, and 0.11% in Precision. For the Massachusetts Building Dataset, PUGNet yields improvements of 0.45% in IoU, 0.30% in F1-score, and 1.46% in Precision. On the Inria Aerial Image Labeling Dataset, our model further achieves gains of 0.59% in IoU, 0.35% in F1-score, 0.67% in Precision, and 0.06% in Recall. These consistent improvements validate the effectiveness of our uncertainty decoupling strategy, where explicit modeling of foreground, background, and contextual uncertainties facilitates more reliable foreground–background separation under complex spatial patterns.

3.5.2. Single-temporal cropland extraction

Table 3 presents the quantitative results for cropland extraction, where PUGNet again achieves the best performance across all metrics. Compared with the previous state-of-the-art DBBANet (Li et al., 2025d), PUGNet improves IoU by 2.62%. It also outperforms UANet (Li et al., 2024b) by 0.99% in F1-score and 0.19% in Precision, and obtains a substantial 2.38% gain in Recall over AAFormer. These results demonstrate

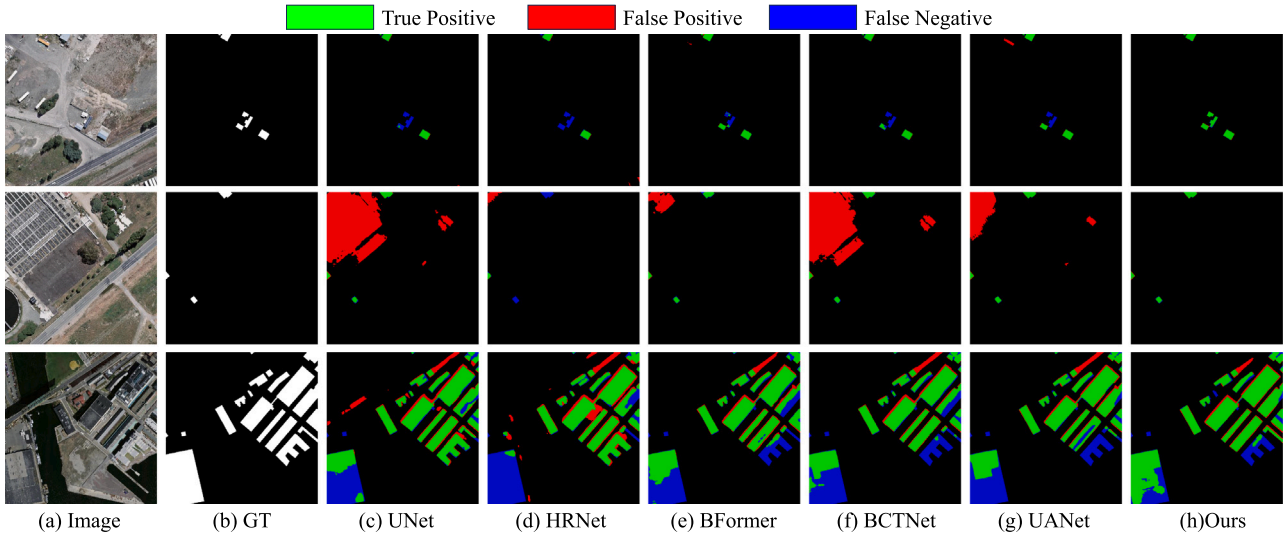


Fig. 3. Visual Comparison on Single-temporal building extraction task. BFormer is an abbreviation for BuildFormer.

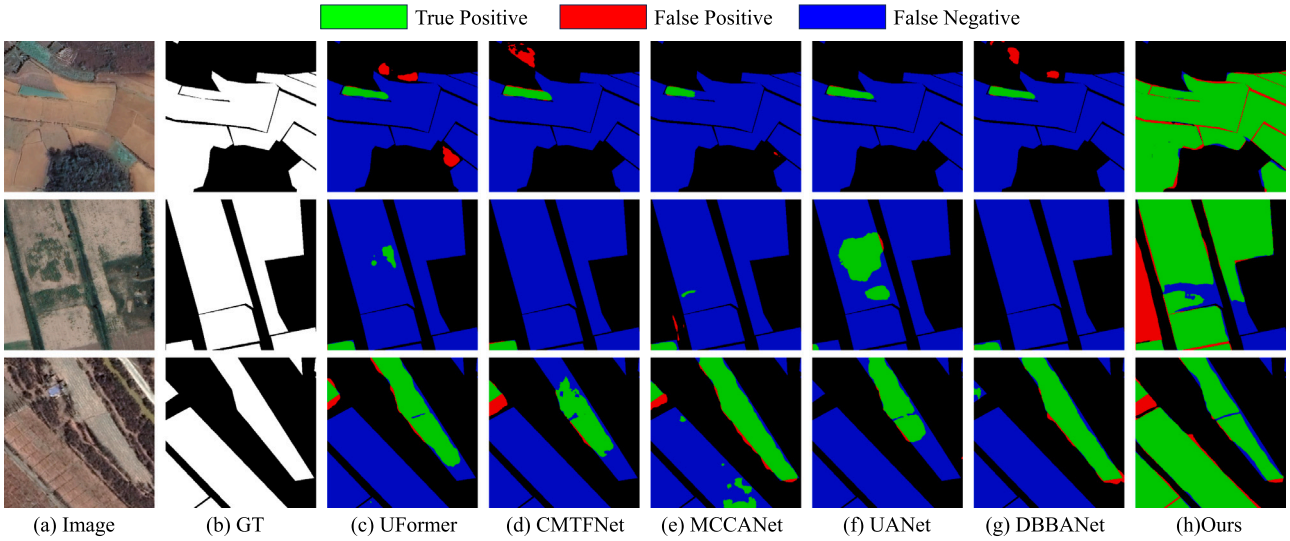


Fig. 4. Visual Comparison on Single-temporal cropland extraction task. UFormer is an abbreviation for UNetFormer.

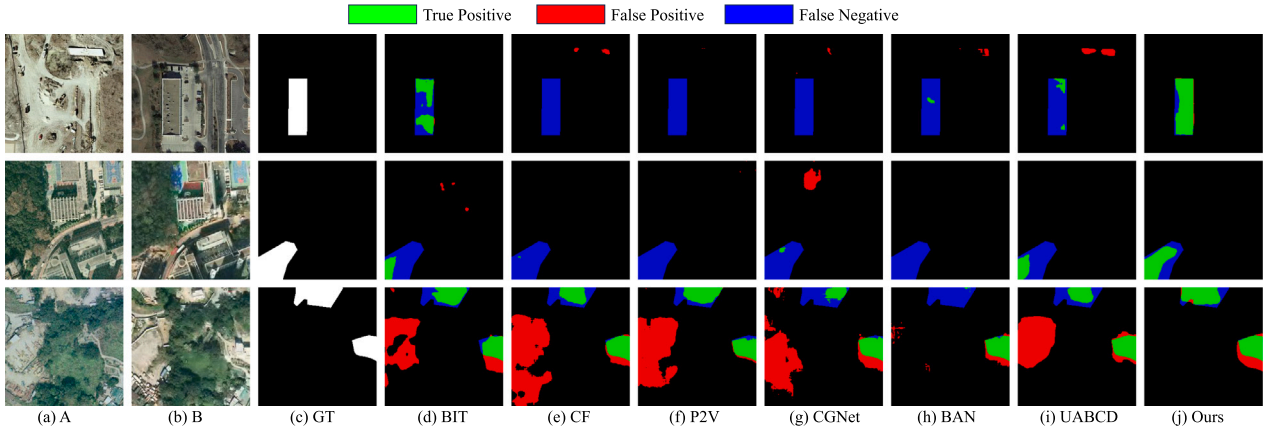


Fig. 5. Visual Comparison on Bi-temporal Building Change Detection task. CF is an abbreviation for ChangeFormer.

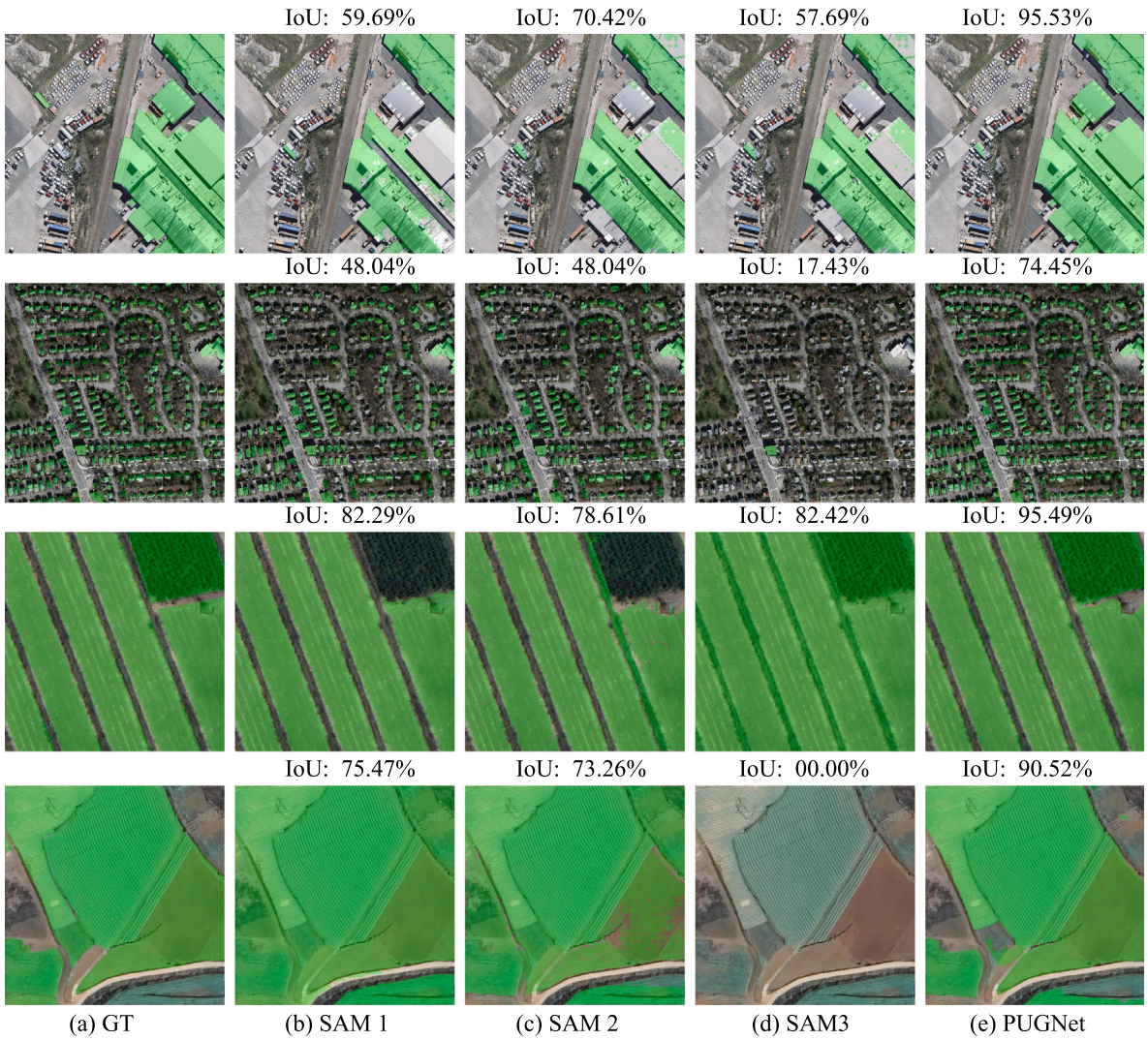


Fig. 6. Visual comparison between SAM-family models and PUGNet on single-temporal building and cropland extraction. For all methods, the ground-truth annotations or predicted masks are overlaid on the original images in green with an opacity of 0.3 to highlight spatial boundaries and structural differences.

that the progressive uncertainty decoupling framework effectively captures heterogeneous textures and large intra-class variations commonly seen in cropland scenes.

3.5.3. Bi-temporal building change detection

Table 4 presents the performance of PUGNet across six building change detection benchmarks. On the LEVIR-CD dataset, PUGNet achieves 86.03% IoU and 92.49% F1-score, performing on par with the best existing methods. On SYSU-CD, PUGNet achieves notable gains of 0.92% IoU and 0.42% Recall, indicating stronger sensitivity to subtle structural changes. On the Lebedev dataset, PUGNet attains 94.00% IoU, substantially surpassing all competing approaches.

Additionally, PUGNet continues to show consistent improvements. On WHU-CD, it increases IoU by 0.73% compared with ChangeMamba and improves the F1-score from 93.69% to 94.10%. On Google-CD, PUGNet outperforms previous methods by 0.75% in IoU and 0.45% in F1-score. On LEVIR-CD+, PUGNet further achieves gains of 0.47% in IoU and 0.22% in F1-score over the strongest baseline. These results reflect the excellent generalization ability of the proposed uncertainty-guided framework, enabling more reliable modeling of temporal inconsistencies and more accurate detection of structural changes.

3.6. Visual evaluation

3.6.1. Single-temporal building extraction

As shown in Fig. 3, we provide three representative visual comparisons. In the first example, buildings occupy a very small portion of the scene, leading to missed detections by most baselines. PUGNet successfully identifies all building instances. In the second example, several methods, including UNet, BCTNet, and UANet, incorrectly classify background regions as buildings, whereas our method produces a prediction closely aligned with the ground truth. In the third example, baseline models fail to extract the small buildings located in the lower-left area of the image, while our method accurately captures their complete shapes. These results highlight the robustness of PUGNet under complex spatial distributions and background interference.

3.6.2. Single-temporal cropland extraction

Fig. 4 illustrates three challenging cropland scenarios. In the first scenario, most models incorrectly segment only green-colored croplands, ignoring large portions of cropland with diverse appearances. PUGNet generates a comprehensive segmentation. In the second scenario, nearly all baselines fail to detect the cropland regions, while

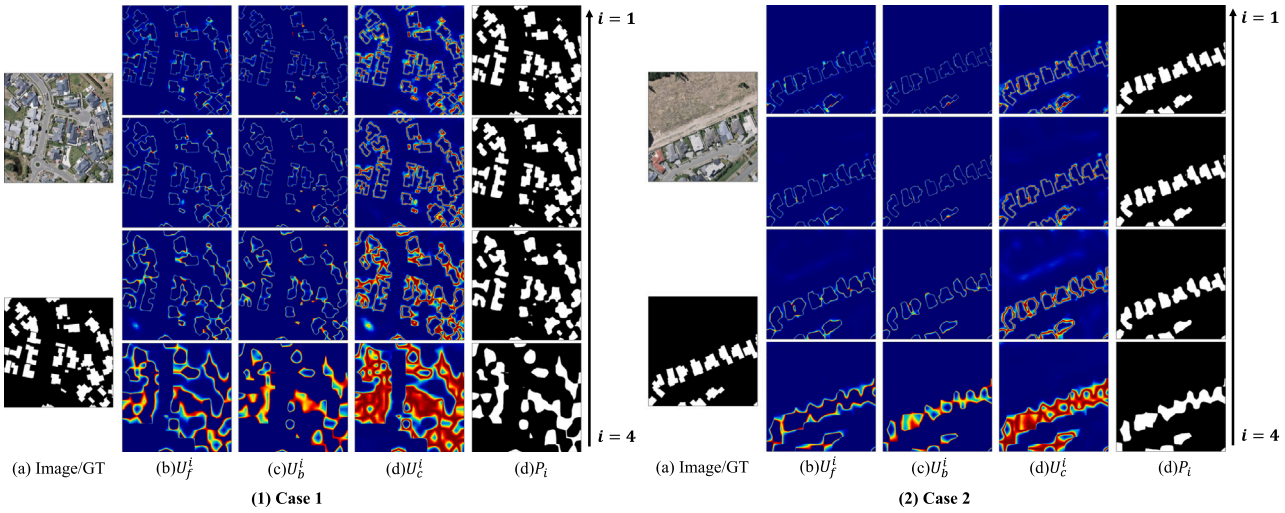


Fig. 7. Visualizations of the progressive uncertainty decomposition and prediction results for the building extraction task.

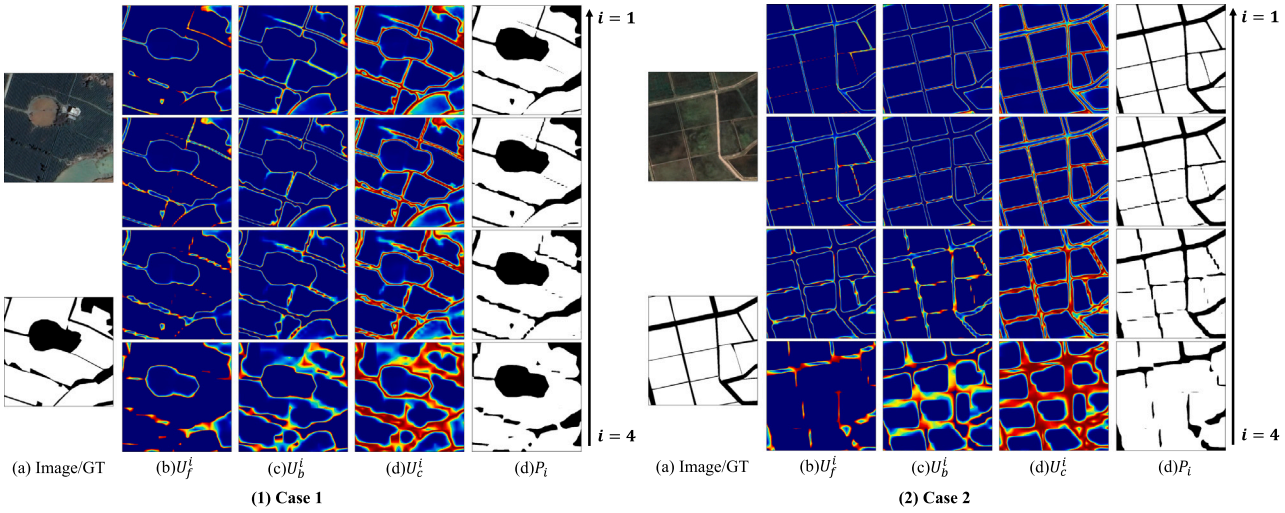


Fig. 8. Visualizations of the progressive uncertainty decomposition and prediction results for the cropland extraction task.

PUGNet successfully captures the primary structures. In the third scenario, significant intra-class variations result in fragmented predictions from other methods, yet PUGNet remains robust and produces accurate delineation. These examples confirm PUGNet's strong adaptability to complex cropland patterns.

3.6.3. Bi-temporal building change detection

Fig. 5 shows three examples from the building change detection task. In the first case, a newly constructed building is overlooked by most baselines except BIT, whereas PUGNet precisely identifies the entire change region. In the second case, only UABCD and PUGNet correctly locate the change, with PUGNet offering the most complete result. In the third case, many models mistakenly treat color variations as structural change, while PUGNet correctly detects the true change areas with minimal false positives. These comparisons demonstrate the strong capability of PUGNet in distinguishing real structural changes from irrelevant variations.

3.7. Comparison with the SAM models

We further provide both quantitative and qualitative comparisons between PUGNet and several SAM-based models for single-temporal building and cropland extraction. As shown in Tables 2 and 3, prompting SAM and SAM2 with PUGNet-generated masks leads to a clear performance drop across all datasets. This reveals that these models cannot effectively refine or preserve fine-scale structures despite receiving high-quality prompts. SAM3, based solely on text prompting, performs even worse; for example, it reaches only 37.04 percent IoU on the Massachusetts dataset and 10.98 percent IoU on cropland extraction. These results highlight the limitations of generic prompt-driven segmentation for fine-grained RS tasks.

The qualitative comparisons in Fig. 6 further support this observation. Even with PUGNet predictions as prompts, the outputs of SAM and SAM2 remain inferior to those of PUGNet, while SAM3, relying only on text prompts, shows the weakest performance. Overall, these comparisons demonstrate that the multi-source, uncertainty-aware design of PUGNet is more effective for RS segmentation than generic prompt-driven models.

4. Ablation study

In the following section, we primarily investigate the impact of uncertainty decomposition, evaluate the effectiveness of the proposed progressive uncertainty-guided strategy, and perform a complexity analysis. To validate the effectiveness of our approach, we conduct experiments on two representative tasks: single-temporal building extraction using the Inria Aerial Image Labeling dataset and bi-temporal building change detection on the LEVIR-CD dataset.

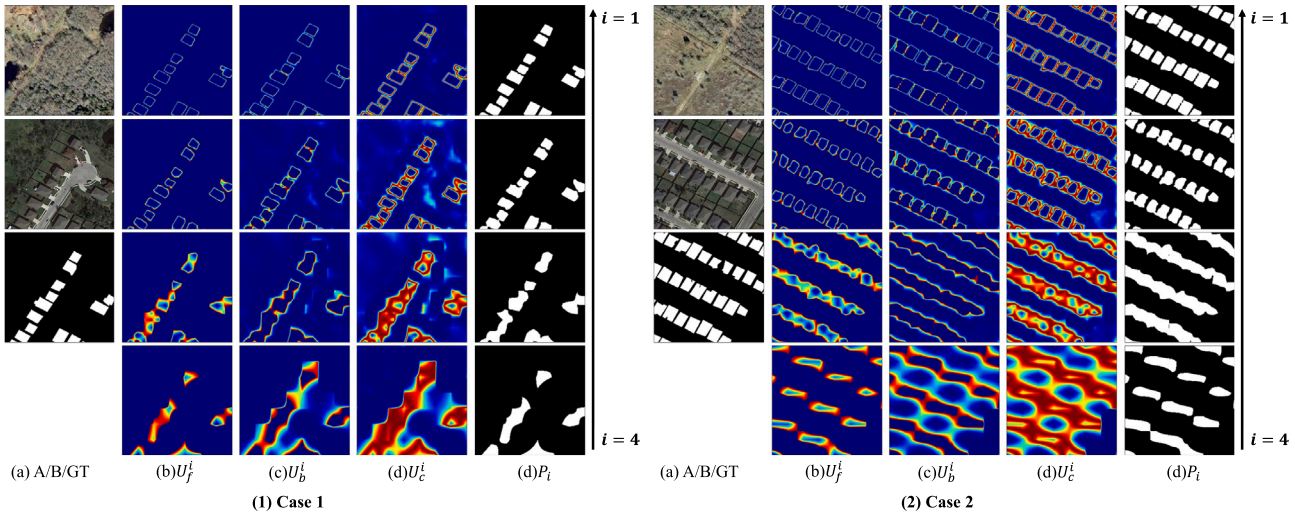


Fig. 9. Visualizations of the progressive uncertainty decomposition and prediction results for the building change detection task.

Table 5
Ablation study results of DUAM and EDRM. The *Base* model corresponds to the network without any uncertainty-aware modules or contrastive loss.

Combination				Inria		LEVIR-CD	
Base	DUAM		EDRM	*IoU* ↑	*F*1 ↑	*IoU* ↑	*F*1 ↑
	fore	back					
✓				80.82	89.45	83.65	91.11
✓	✓			81.95	90.06	84.75	91.74
✓	✓	✓		82.43	90.24	84.93	91.80
✓			✓	82.81	90.60	85.33	92.08
✓	✓	✓	✓	**83.93**	**91.26**	**86.03**	**92.49**

4.1. Why three types of uncertainty?

As shown in Table 5, the progressive incorporation of three distinct uncertainty components, namely foreground uncertainty (DUAM, fore), background uncertainty (DUAM, back), and contextual uncertainty (EDRM), each yields significant improvements in segmentation performance. This explicit decoupling and modeling of heterogeneous uncertainties collectively result in substantial overall gains, highlighting the effectiveness of our approach. Fig. 7, Fig. 8, and Fig. 9 further illustrate that these three types of uncertainty focus on different spatial regions. Foreground uncertainty is primarily focused on the challenging-to-sample areas around the foreground objects, while background uncertainty is closely associated with the edges. Contextual uncertainty, on the other hand, emphasizes the global consistency of the segmentation, reflecting the instability of the overall structure. This decoupling strategy offers a key advantage over traditional uncertainty approaches, which typically focus solely on global uncertainty modeling. By considering both local (foreground and background) and global (contextual) uncertainties, our method captures the dynamic relationship between foreground and background at the local level while also maintaining the global stability and consistency.

4.2. Why the progressive guided strategy?

As discussed earlier, many existing uncertainty methods rely on explicit uncertainty-estimation modules, which add design complexity. Our approach instead adopts a progressive decoupling strategy that directly derives foreground, background, and contextual uncertainty from coarse predictions. This provides clear semantic cues for refinement and enables the network to gradually resolve ambiguous regions in a structured manner.

The visualizations in Fig. 7, Fig. 8, and Fig. 9 illustrate the effectiveness of this strategy across single-temporal building extraction, cropland extraction, and bi-temporal building change detection. In the initial stage, uncertainties are highly entangled and coarse predictions show confusion between object and background. As progressive guidance proceeds, the uncertainty maps become more organized, with foreground, background, and contextual regions clearly separated. This improvement is reflected in higher-stage predictions, where object boundaries and structural details are better preserved. Along with the quantitative results in Table 6, these visualizations confirm the effectiveness of our uncertainty-guided progressive refinement mechanism.

4.3. Complexity analysis

As shown in Fig. 10, we compare PUGNet with representative single- and bi-temporal RS binary segmentation methods in terms of computational complexity, parameter count, and inference speed (FPS). For single-temporal segmentation, PUGNet achieves a balanced trade-off with 60.0 GFLOPs, 63.4M parameters, and 35.0 FPS, offering higher representational capacity than lightweight models such as UNetFormer (11.7 GFLOPs, 11.7M, 129.9 FPS) while being more efficient than heavyweight models like UNet (215.1 GFLOPs, 147.8M, 51.0 FPS) and DSNet (210.5 GFLOPs, 26.1M, 19.8 FPS).

In bi-temporal change detection, PUGNet maintains competitive efficiency (28.0 GFLOPs, 64.2M, 42.0 FPS) and outperforms many existing methods in overall balance. It achieves better inference speed than the RS Mamba (18.4 GFLOPs, 42.3M, 15.6 FPS) and Change Mamba (44.8 GFLOPs, 85.5M, 15.5 FPS) while providing higher representational capacity than extremely lightweight models such as BIT (8.5 GFLOPs, 3.0M, 131.2 FPS) or HANet (17.6 GFLOPs, 2.6M, 96.2 FPS), and remains more efficient than complex models like ChangeFormer (202.8 GFLOPs, 41.0M, 36.4 FPS) and M-Swin (44.2 GFLOPs, 126.4M, 18.0 FPS).

5. Parametric analysis

5.1. Settings of centers in DUAM

As described in the methodology section, we introduce learnable centroids to quantify foreground and background uncertainty levels via the Gaussian soft ranking algorithm, empirically setting the number of centroids to five. Notably, the Gaussian soft ranking does not perform explicit sorting; instead, it uses differentiable Gaussian kernels to approximate relative ranks. While this mapping is non-linear, its

Table 6
Quantitative evaluation at each level (P_i) of our PUGNet.

Output	Inria				LEVIR-CD			
	IoU ↑	*F1* ↑	*Pre* ↑	*Recall* ↑	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
P_4	73.30	84.60	86.47	82.80	67.29	80.45	81.47	79.46
P_3	81.24	89.65	90.53	88.79	79.26	88.43	87.66	89.22
P_2	83.41	90.95	92.23	89.71	84.65	91.69	92.85	90.56
P_1	**83.93**	**91.26**	**92.53**	**90.03**	**86.03**	**92.49**	**93.59**	**91.42**

Table 7
Quantitative evaluation of PUGNet performance with varying scale factors. Here, Ours means $\{\frac{1}{2}, \frac{3}{4}, 1, \frac{5}{4}, \frac{3}{2}\}$.

Factors	Inria				LEVIR-CD			
	IoU ↑	*F1* ↑	*Pre* ↑	*Recall* ↑	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
$\{1\}$	82.55	90.40	91.72	89.11	84.23	91.61	92.35	90.88
$\{\frac{1}{2}, \frac{3}{4}, 1\}$	83.28	90.71	91.79	89.65	85.67	92.19	93.26	91.14
$\{1, \frac{5}{4}, \frac{3}{2}\}$	83.45	90.86	92.05	89.71	85.82	92.30	93.39	91.24
$\{\frac{1}{2}, 1, \frac{3}{2}\}$	83.52	90.85	92.33	89.42	85.83	92.33	93.43	91.25
Ours	**83.93**	**91.26**	**92.53**	**90.03**	**86.03**	**92.49**	**93.59**	**91.42**

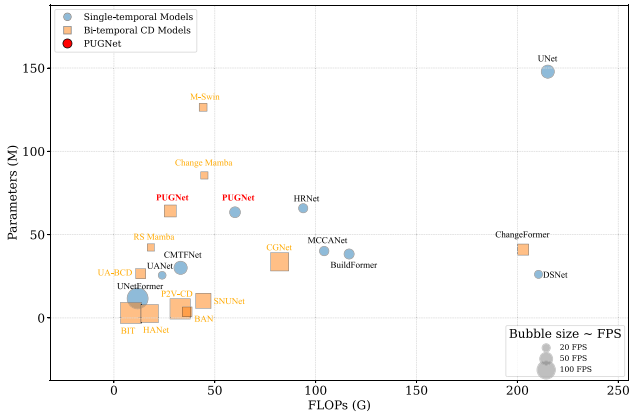


Fig. 10. Comparison of model complexity, parameter count, and inference speed for single- and bi-temporal RS binary segmentation. FLOPs (G) are shown on the x-axis, parameters (M) on the y-axis, and bubble size corresponds to FPS. Circles indicate single-temporal models (input size 512 × 512), squares indicate bi-temporal models (input size 256 × 256), and PUGNet (backbone: PVT-V2) is highlighted in red. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

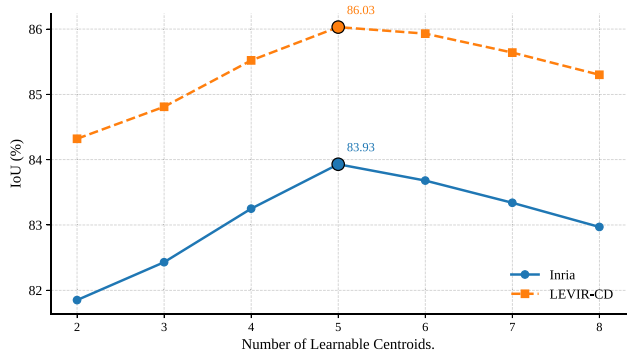


Fig. 11. Quantitative comparison of PUGNet with different numbers of learnable centroids.

computational cost scales linearly with the number of centroids and image size. To systematically evaluate the effect of centroid quantity, we conducted ablation experiments with 2 to 8 centroids (2, 3, 4, 5, 6, 7, 8), with results shown in Fig. 11.

Table 8
Quantitative comparison of PUGNet performance with and without contrastive learning strategy.

CL	Inria				LEVIR-CD			
	IoU ↑	*F1* ↑	*Pre* ↑	*Recall* ↑	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
✗	83.75	90.38	91.75	89.06	85.88	92.34	93.45	91.26
✓	**83.93**	**91.26**	**92.53**	**90.03**	**86.03**	**92.49**	**93.59**	**91.42**

Our analysis reveals that when the number of centroids is insufficient (2, 3, 4), the model exhibits relatively poor performance, suggesting that fewer uncertainty categories fail to adequately capture the full spectrum of uncertainty variations. Conversely, when employing a larger number of centroids (6, 7, 8), the model demonstrates substantial improvement compared to configurations with fewer centroids, although performance remains slightly inferior to the optimal setting of 5 centroids. This observation indicates that increased centroid numbers enable the model to better discriminate uncertainty variations across different spatial regions, thereby facilitating more effective attention allocation to high-uncertainty areas. However, when the centroid count becomes excessive, overly fine-grained uncertainty categorization can diminish inter-regional correlation and coherence, consequently leading to a measurable degree of performance degradation. This phenomenon suggests that there exists an optimal balance between uncertainty granularity and model effectiveness, with 5 centroids representing the most favorable configuration for our experimental settings.

Overall, these results support our empirical choice of $K = 5$ centroids as providing the best balance between uncertainty granularity, feature discriminability, and model effectiveness for our experimental settings.

5.2. Contribution of the contrastive learning

In DUAM, we incorporate a contrastive learning strategy to explicitly enlarge the feature separation between foreground and background, thereby enhancing their discriminative representations. To evaluate its effectiveness, we conduct an ablation study by removing the contrastive term and retaining only the concatenation operation followed by a 1×1 convolution for feature fusion. As shown in Table 8, discarding the contrastive learning strategy leads to a noticeable performance drop, confirming its importance for robust feature refinement. This observation consistently demonstrate that the contrastive learning strategy effectively strengthens feature discriminability and contributes to improved segmentation performance.

5.3. Scale factors of EDRM

To explore global contextual relevance at different scales, we examine the uncertainty represented by entropy values across multiple scale spaces and enhance the features accordingly. In our approach, we select the scale factors as $\{\frac{1}{2}, \frac{3}{4}, 1, \frac{5}{4}, \frac{3}{2}\}$. To evaluate the impact of different scale factor choices on experimental outcomes, we conducted several comparative experiments with scale factor sets $\{\frac{1}{2}, \frac{3}{4}, 1\}$, $\{1, \frac{5}{4}, \frac{3}{2}\}$, $\{\frac{1}{2}, 1, \frac{3}{2}\}$, and $\{1\}$. The results are presented in Table 7. As shown, compared to not exploring global contextual relevance at multiple scales, incorporating multi-scale exploration yields a significant improvement in performance. Furthermore, the combined scaling approach, rather than purely enlarging or shrinking, leads to more noticeable gains. However, excessive scaling results in a higher computational cost. Therefore, in our PUGNet, we adopt the scale space of $\{\frac{1}{2}, \frac{3}{4}, 1, \frac{5}{4}, \frac{3}{2}\}$.

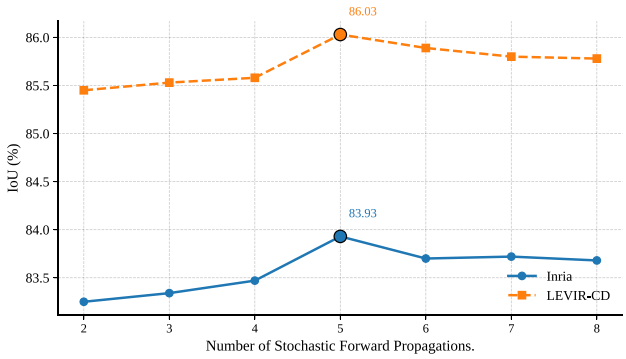


Fig. 12. Quantitative comparison of PUGNet performance with varying numbers of stochastic forward propagations.

Table 9
Quantitative comparison of PUGNet performance with/without entropy weighting (EW) and Monte Carlo (MC) sampling.

EW	MC	Inria				LEVIR-CD			
		IoU ↑	*F1* ↑	*Pre* ↑	*Recall* ↑	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
✗	✗	82.73	90.59	91.94	89.28	85.27	91.99	93.02	90.99
✗	✓	83.14	90.70	91.69	89.74	85.48	92.19	93.12	91.27
✓	✓	**83.93**	**91.26**	**92.53**	**90.03**	**86.03**	**92.49**	**93.59**	**91.42**

5.4. Contribution of Monte Carlo sampling

In our proposed EDRM, we introduce a Monte Carlo-aware refinement module that focuses on high-entropy pixels to capture the inherent variability in ambiguous regions. By sampling the input features N times and computing both the mean prediction and variance across these samples, the module enhances feature representations and leverages uncertainty to refine the model's output in critical regions such as boundaries and mixed pixels.

To quantify the impact of both entropy weighting (EW) and Monte Carlo (MC) sampling, we conduct an ablation study comparing model performance under different configurations, as shown in Table 9. Without either EW or MC, the model achieves an IoU of 82.73% and F1 of 90.59% on Inria, and 85.27% and 91.99% on LEVIR-CD. Introducing MC sampling alone improves performance to 83.14% IoU and 90.70% F1 on Inria, and 85.48% IoU and 92.19% F1 on LEVIR-CD, confirming the effectiveness of targeted Monte Carlo refinement in high-entropy regions. Finally, combining both EW and MC yields the best results, with IoU/F1 increasing to 83.93%/91.26% on Inria and 86.03%/92.49% on LEVIR-CD. These results demonstrate that EW and MC provide complementary benefits: EW emphasizes pixels with high semantic ambiguity, while MC sampling captures feature variability in these regions, together enhancing feature stability, mitigating contextual uncertainty, and improving segmentation accuracy.

To determine an appropriate sampling size, we evaluated the model under different numbers of stochastic forward passes. As shown in Fig. 12, using $N = 5$ achieves the best performance, providing a good balance between uncertainty estimation quality and computational cost. When N becomes larger than 5, the performance gains quickly saturate and may even slightly decline. This phenomenon is consistent with previous observations in uncertainty estimation literature, where excessive Monte Carlo sampling amplifies the averaging effect and leads to over-smoothed feature representations and weakened boundary details (Kendall and Gal, 2017; Gal and Ghahramani, 2016; Cai et al., 2024). Moreover, sampling with a scale much larger than that used during training introduces a mild distribution mismatch, which further affects prediction accuracy. Therefore, we adopt $N = 5$ as the optimal choice for our framework.

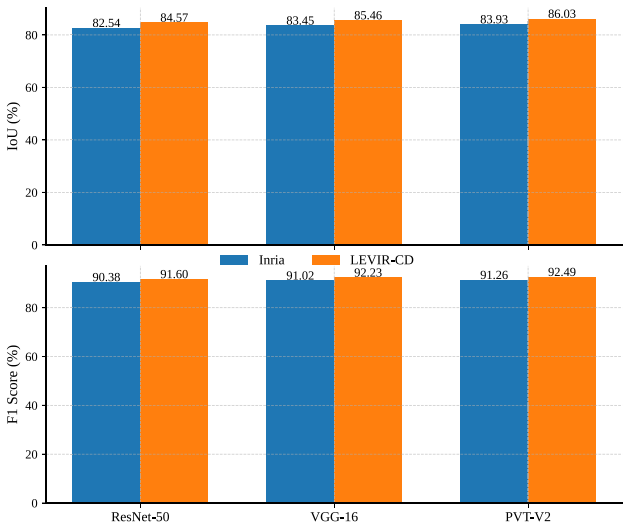


Fig. 13. Performance comparison of PUGNet with different backbones (CNN-based: ResNet-50, VGG-16; Transformer-based: PVTv2) on building extraction (Inria) and building change detection (LEVIR-CD) tasks.

5.5. Impact of backbones

As illustrated in Fig. 13, we conduct comprehensive performance evaluation across various backbone architectures on both building extraction and building change detection tasks. The experimental results clearly demonstrate that transformer-based encoders consistently achieve superior performance compared to their CNN-based counterparts, indicating the effectiveness of self-attention mechanisms in capturing long-range spatial dependencies for RS segmentation tasks. Notably, regardless of the specific backbone architecture employed, our PUGNet consistently delivers excellent performance across diverse segmentation scenarios, with performance improvements observed across all tested configurations. This consistent superiority across different architectural foundations conclusively validates the effectiveness of our progressive uncertainty-guided decoding strategy, highlighting its ability to enhance segmentation performance independent of the underlying feature extraction mechanism.

5.6. Sensitivity analysis

To evaluate the robustness of PUGNet with respect to uncertainty-related hyperparameters, we conducted a sensitivity analysis on the Inria and LEVIR-CD datasets. We varied the Gaussian smoothness parameter β in DUAM, the temperature parameter τ in the contrastive loss, and the multi-scale kernel configuration $\{k_j\}$ in EDRM, while keeping all other settings unchanged. As shown in Fig. 14, both IoU and F1-score vary smoothly within a narrow range across all tested settings, and the overall performance consistently remains superior to the baseline. These results indicate that PUGNet is not sensitive to precise hyperparameter tuning, and that the observed performance gains stem from the principled region-aware uncertainty decomposition rather than fine-grained parameter tuning.

5.7. Comparison with previous uncertainty strategies

To further validate the effectiveness of our foreground, background, and context decoupled uncertainty modeling, we conduct a comprehensive comparison against several commonly used uncertainty quantification strategies. Specifically, we consider four representative alternatives: (1) the uncertainty ranking algorithm (URA) from UANet (Li et al., 2024b), (2) the threshold-based sample selection strategy from

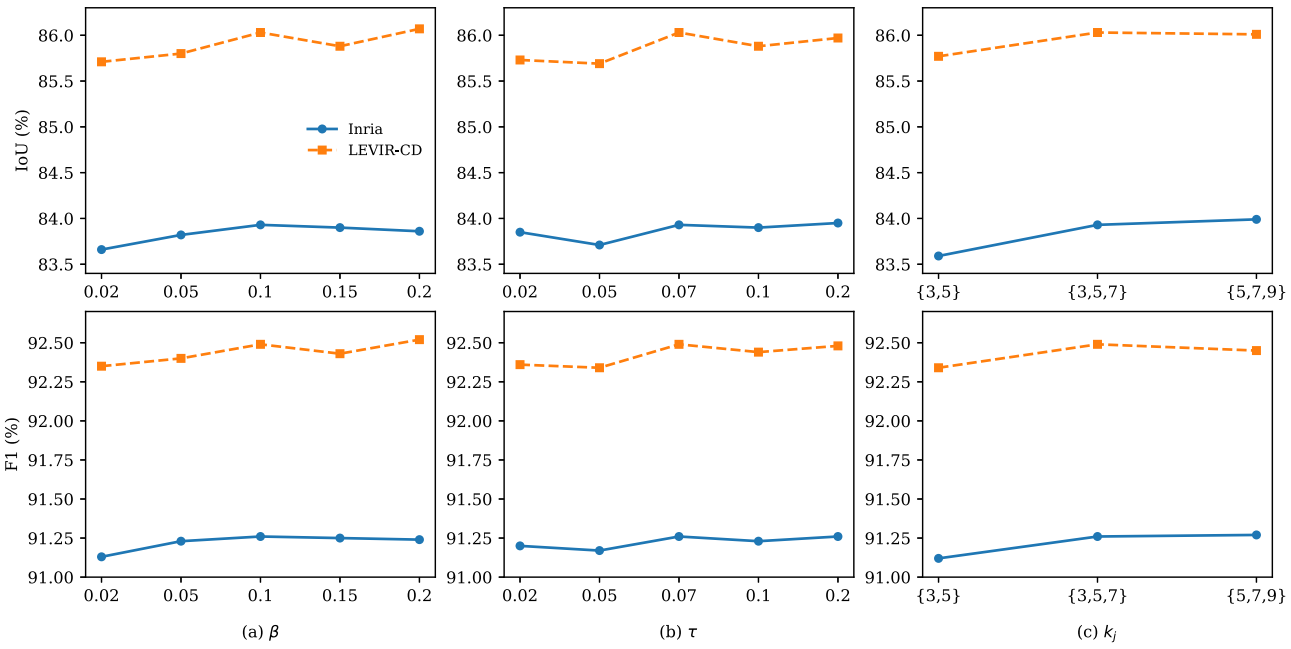


Fig. 14. Sensitivity analysis of key hyperparameters in PUGNet. The first row reports IoU, and the second row reports F1-score. (a) Gaussian smoothness parameter β in DUAM, (b) temperature parameter τ in the contrastive loss, and (c) kernel set k_j in the entropy-driven refinement module. Results are shown on the Inria and LEVIR-CD datasets.



Fig. 15. Comparison of uncertainty estimation between the proposed PUGNet and other uncertainty-aware methods for both single-temporal building extraction and bi-temporal building change detection under large-scale scenarios. Note that *bi* denotes the variant adapted for bi-temporal segmentation.

OHEM loss (Shrivastava et al., 2016) with a threshold of 0.7, (3) entropy-based uncertainty weighting (Landgraf et al., 2023; da Cruz et al., 2024), and (4) the CVAE-based aleatoric uncertainty estimation and GAN-based epistemic uncertainty modeling adopted in UABCD (Li et al., 2025a) and PUENet (Zhang et al., 2023), referred to as CVAE+GAN.

Table 10
Quantitative comparison of PUGNet performance with different uncertainty quantification (UQ) methods. 'CG' denotes the original CVAE+GAN method for brevity.

UQ	Inria				LEVIR-CD			
	IoU ↑	*F1* ↑	*Pre* ↑	*Recall* ↑	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
✗	80.82	89.45	89.82	89.09	83.65	91.11	92.13	90.11
URA	83.47	90.74	92.03	89.49	85.38	92.19	93.17	91.24
OHEM	81.54	89.96	90.91	89.02	83.72	91.11	92.14	90.11
Entropy	82.26	89.60	89.96	89.25	84.65	91.79	92.83	90.78
CG	83.60	91.09	92.41	89.81	85.95	92.40	93.32	**91.50**
Ours	**83.93**	**91.26**	**92.53**	**90.03**	**86.03**	**92.49**	**93.59**	91.42

For a fair comparison, each uncertainty estimation strategy is directly integrated into our baseline model, which uses PVT-V2 as the encoder and incorporates RFB modules together with a U-shaped progressive decoder. All other configurations remain unchanged. As presented in Table 10, we observe that all existing uncertainty estimation strategies bring noticeable performance improvements. Among them, OHEM achieves the smallest gain, mainly because its fixed threshold mechanism is too coarse for fine-grained remote sensing segmentation. In contrast, our decoupled uncertainty estimation achieves the most significant improvement across datasets. These findings confirm that explicitly separating foreground, background, and context uncertainties is an effective design that substantially enhances segmentation performance.

As shown in Fig. 15, we evaluate PUGNet on two challenging large-scale tasks: single-temporal building extraction and bi-temporal building change detection, using the WHU Building Change Detection Dataset (Ji et al., 2018a). To handle the large spatial extent of the test images, we adopt a sliding-window inference strategy with multi-batch prediction to ensure both efficiency and complete spatial coverage. We compare the uncertainty estimation performance of PUGNet with representative uncertainty-aware baselines, including UDNet (Fang et al.,

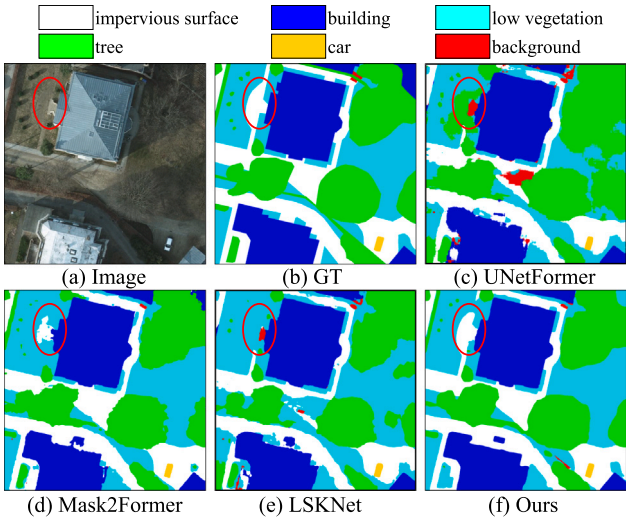


Fig. 16. Visual comparison of semantic segmentation performance across different methods on the ISPRS Potsdam multi-class RS dataset.

2023), UANet (Li et al., 2024b), and PUENet (Zhang et al., 2023). For bi-temporal tasks, where UDNet and PUENet were originally designed for single-temporal RGB segmentation, we extend them with a Siamese encoder structure, followed by channel-wise concatenation and a 1×1 convolution for temporal adaptation.

The entropy maps in Fig. 15 clearly show that PUGNet yields the lowest and most stable uncertainty distributions, with significantly fewer high-entropy regions (*i.e.*, uncertain predictions). This indicates that PUGNet effectively suppresses pixel-level ambiguity and produces more confident, consistent segmentation results. Furthermore, under large-scale inference, our model demonstrates superior segmentation completeness and more regular, well-aligned boundaries in both building extraction and change detection tasks, highlighting its strong potential for real-world RS applications.

6. Discussion

6.1. Extension to multi-class RS segmentation

To assess the generalizability of our approach to multi-class segmentation tasks, we extended our method for multi-class RS semantic segmentation. For each class, we treat it as the foreground and quantify foreground, background, and contextual uncertainties based on the corresponding probability map in the coarse segmentation output. In DUAM, the same processing steps from the binary segmentation setting are applied to each class individually, producing class-specific uncertainty-enhanced features for foreground and background, which are subsequently fused for final prediction.

Experiments were conducted on the Potsdam multi-class semantic segmentation dataset following the same settings as in Li et al. (2025c). We compared our method with three CNN-based approaches (DeeplabV3+ Chen et al., 2018, ResUNet Diakogiannis et al., 2020, ABCNet Li et al., 2021) and four Transformer-based methods (BANet Wang et al., 2021a, UNetFormer Wang et al., 2022b, Mask2Former Cheng et al., 2022, LSKNet Li et al., 2025c). Evaluation metrics included mIoU, mF1, and overall accuracy (OA). As shown in Table 11, our method consistently achieves competitive or superior performance in both mIoU and mF1, demonstrating the scalability and effectiveness of our uncertainty decoupling strategy for multi-class segmentation. Moreover, as illustrated in Fig. 16, PUGNet produces clearer and more accurate visual segmentation results compared with existing methods.

Table 11
Performance comparison of single-temporal semantic segmentation with baseline models on the Postdam dataset. ↑ indicates that higher scores are better. The best score for each metric is marked in bold.

Baseline	$mIoU$ ↑	$mF1$ ↑	OA ↑
CNN-based backbone			
DeeplabV3+	84.3	92.1	90.8
ResUNet	86.9	92.9	91.3
ABCNet	86.5	92.7	91.3
Transformer-based backbone			
BANet	86.3	92.5	91.0
UNetFormer	86.8	92.9	91.7
Mask2Former	86.6	92.5	**92.5**
LSKNet	86.7	92.9	91.7
PUGNet	**87.1**	**93.0**	91.5

6.2. Limitation and future work

While PUGNet demonstrates strong performance for binary semantic segmentation in RS images, it still has several limitations. The proposed progressive decoding framework is built upon a tripartite uncertainty decomposition into foreground, background, and contextual components. As a result, its effectiveness relies on a sufficient degree of distinction between foreground and background regions. When the spatial resolution is low or when foreground and background exhibit high visual similarity, such as cropland areas covered by different crop types, the model may produce noticeable errors, as illustrated in Fig. 17.

Moreover, the current framework is primarily designed for binary segmentation tasks. Extending this uncertainty decomposition to multi-class semantic segmentation introduces additional challenges, particularly in scenarios where certain categories are extremely rare. In such cases, uncertainty estimation itself may become biased due to insufficient observations, which can undermine the reliability of class-wise uncertainty modeling and necessitate more sophisticated regulation or re-balancing mechanisms. This issue is especially pronounced in remote sensing datasets with highly imbalanced class distributions. Despite these challenges, the proposed uncertainty-aware decoupling strategy remains effective and well-justified for binary segmentation problems, where foreground–background separation constitutes the dominant source of ambiguity. By explicitly modeling foreground, background, and contextual uncertainty, PUGNet is able to alleviate boundary confusion and enhance structural consistency, as validated by extensive experiments on multiple benchmark datasets.

Addressing the above limitations motivates several directions for future work. First, we aim to enhance robustness under low-resolution or low-contrast conditions by incorporating richer multi-scale representations and more advanced context modeling techniques. Second, we plan to explore a more general multi-class uncertainty decomposition framework that can dynamically adapt to varying class frequencies and inter-class similarities, thereby extending the applicability of PUGNet to more complex remote sensing segmentation scenarios.

7. Conclusion

In this paper, we propose a Progressive Uncertainty-Guided Segmentation Network (PUGNet) for RS binary segmentation. PUGNet decouples the uncertainty in coarse segmentation predictions derived from high-level features into three distinct components: foreground uncertainty, background uncertainty, and contextual uncertainty. Foreground and background uncertainties model the dynamic local relationships between objects and their surroundings, while contextual uncertainty captures the overall consistency of the segmentation map. To effectively handle each type of uncertainty, we design corresponding

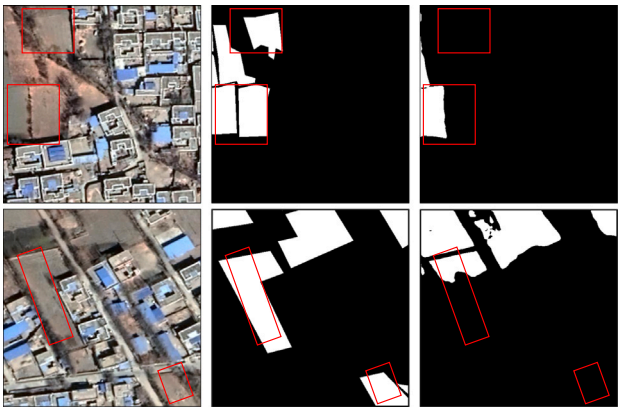


Fig. 17. Illustration of model limitations. When the foreground and background are highly similar, as in cropland areas with different crops, the foreground-background uncertainty decoupling may fail, leading to significant segmentation errors.

uncertainty-aware enhancement strategies. Extensive experiments conducted on single-temporal building extraction, single-temporal cropland extraction, and bi-temporal building change detection tasks clearly demonstrate the effectiveness and superiority of our proposed PUGNet.

CRediT authorship contribution statement

Jiepan Li: Writing – original draft, Software, Resources, Methodology, Data curation. **Wei He:** Writing – review & editing, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Ting Hu:** Writing – review & editing, Supervision, Conceptualization. **Minghao Tang:** Writing – review & editing, Resources, Formal analysis. **Liangpei Zhang:** Writing – review & editing, Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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